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1. The Basic HTTP GET/response interaction

Answer the following Questions based on the trace file: *http-wireshark-trace1-1*.

1. Is your browser running HTTP version 1.0, 1.1, or 2? What version of HTTP is the server running?

Ans : Browser is running HTTP version 1.1 and server is running 1.1 version of HTTP

2. What languages (if any) does your browser indicate that it can accept to the server?

Ans : en-US

3. What is the IP address of your computer? What is the IP address of the gaia.cs.umass.edu server?

IP address of computer : 10.0.0.44

IP address of gaia.cs.umass.edu.server : 128.119.245.12

4. What is the status code returned from the server to your browser?

200 status code was returned from server to browser

5. When was the HTML file that you are retrieving last modified at the server?

Sat, 30 Jan 2021 , 6:59:02

6. How many bytes of content are being returned to your browser?

128 bytes has returned to browser.

7. By inspecting the raw data in the packet content window, do you see any headers within the data that are not displayed in the packet-listing window? If so, name one.

None

2. The HTTP CONDITIONAL GET/response interaction

Answer the following questions based on the trace file *http-wireshark-trace2-1*.

1. Inspect the contents of the first HTTP GET request from your browser to the server. Do you see an "IF-MODIFIED-SINCE" line in the HTTP GET?

No, in few HTTP GET requests "IF-MODIFIED-SINCE" is not present

2. Inspect the contents of the server response. Did the server explicitly return the contents of the file? How can you tell?

Yes — server did return the file, by checking the content-length

3. Now inspect the contents of the second HTTP GET request from your browser to the server. Do you see an "IF-MODIFIED-SINCE:" line in the HTTP GET¹? If so, what information follows the "IF-MODIFIED-SINCE:" header?

Yes, in a few HTTP GET requests "IF-MODIFIED-SINCE" is present. It tells "Only send me the resource if it has been modified **after** this date/time."

4. What is the HTTP status code and phrase returned from the server in response to this second HTTP GET? Did the server explicitly return the contents of the file? Explain.

Status Code : 304

Status description/phrase : Not Modified

No server did not return the contents of the file, we can check this from content-length, where it is absent in this HTTP GET response.

3. Retrieving Long Documents

Answer the following questions based on the trace file: *http-wireshark-trace3-1*.

1. How many HTTP GET request messages did your browser send? Which packet number in the trace contains the GET message for the Bill of Rights?

Only one HTTP request message my browser sent. Packet number :26

2. Which packet number in the trace contains the status code and phrase associated with the response to the HTTP GET request?

Packet number : 32 contains the status code and phrase associated with the response to HTTP GET request

3. What is the status code and phrase in the response?

status code: 200

Phrase: OK

4. How many data-containing TCP segments were needed to carry the single HTTP response and the text of the Bill of Rights?

4 reassembled TCP segments were needed to carry the single HTTP response

4. HTML Documents with Embedded Objects

Answer the following questions trace file *http-wireshark-trace4-1*.

1. How many HTTP GET request messages did your browser send? To which Internet addresses were these GET requests sent?

4 HTTP get request messages. 128.119.245.12, 178.79.137.164, 104.98.115.146

2. Can you tell whether your browser downloaded the two images serially, or whether they were downloaded from the two web sites in parallel? Explain.

They were downloaded from the two web sites in Parallel. They have different IP address request sent.

5 HTTP Authentication

Answer the following questions trace file http-wireshark-trace5-1

1. What is the server's response (status code and phrase) in response to the initial HTTP GET message from your browser?

Status code: 401 , phrase: Unauthorized

2. When your browser's sends the HTTP GET message for the second time, what new field is included in the HTTP GET message?

Authorization Header